

Introduction to Geology

GS 21/L: Physical and Structural Geology

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Definition of Geology

Geology is the fundamental science that pursues the study of the Earth, its dynamic processes, rich history, and intricate physical structures.

It seeks to understand the planet's vast composition and the powerful forces continuously shaping its surface over vast periods of time.



Branches of Geology

Chemical & Physical Properties

Geochemistry

Focuses on the chemical composition and changes of materials that make up the Earth. It analyzes the complex distribution of chemical elements in rocks and minerals to profoundly understand fundamental Earth processes.

Geophysics

Applies principles of physics to study the Earth. It involves measuring seismic waves, gravity, and magnetic fields to investigate the planet's hidden internal structure and massive dynamic tectonic forces.

History & Surface Processes

- **Paleontology:** The study of ancient life through fossils. It reveals the extraordinary history of biological evolution and ancient environments preserved within sedimentary rocks over millions of years.
- **Sedimentology:** Focuses on the precise processes of erosion, transport, deposition, and lithification. It helps geologists accurately interpret past environmental conditions based on distinct sedimentary strata.



Earth's Diverse Systems



Oceanography

The comprehensive study of the marine environment, encompassing ocean currents, vast marine ecosystems, and seafloor geology.



Volcanology

The detailed study of volcanoes, lava, magma, and related geological phenomena, heavily focusing on eruptive mechanisms and hazards.



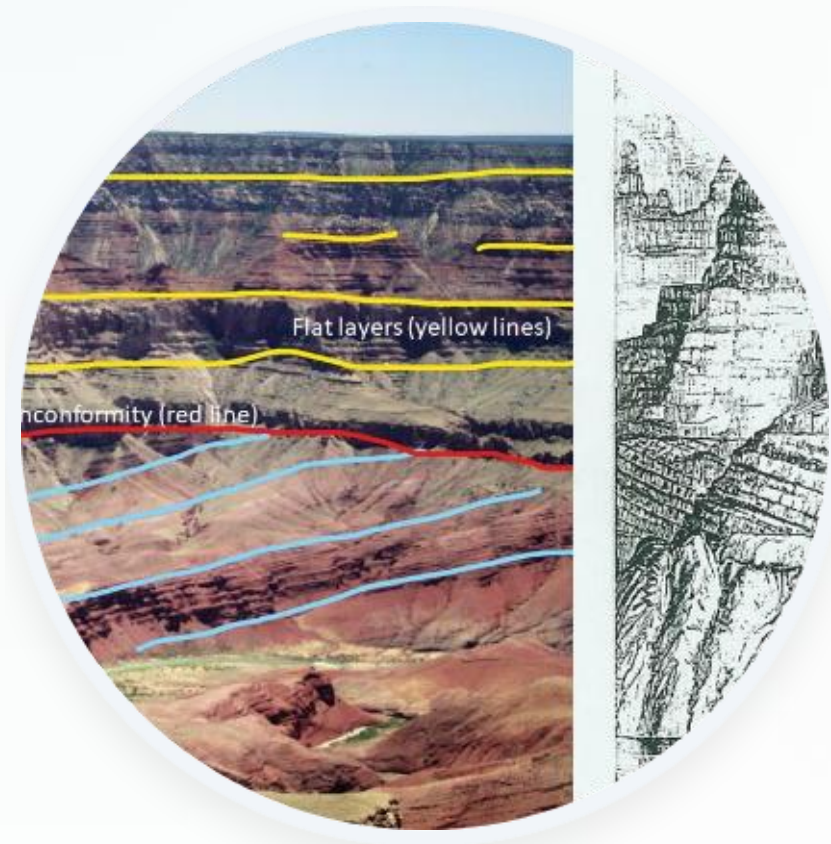
Hydrogeology

Examines the critical distribution, movement, and quality of groundwater circulating in the soil and rocks of the Earth's crust.

Physical vs. Historical Geology

The two major subdivisions that form the foundation of
comprehensive geologic study

Physical Geology



Understanding Earth's Dynamics

Physical geology is the study of the materials composing the Earth and seeks to deeply understand the numerous processes that operate beneath and upon its surface.

It focuses on the mechanisms of rock formation, powerful earthquakes, tectonic shifts, and erosional forces that continually reshape the planet's exterior.

Historical Geology



Tracing Evolutionary Timelines

Historical geology aims to intimately understand the origin of Earth and its breathtaking development through time.

By utilizing the principles of physical geology, it establishes a chronological arrangement of the multitude of physical and biological changes that have occurred in the geological past, utilizing fossils as key indicators.

Early Views vs. Modern Geology

Catastrophism

Proposed in the mid-1600s by James Ussher, this view created a rigid chronology that placed Earth's creation exactly at 4004 B.C.

It theorized that the Earth's landscape was shaped primarily by sudden, great catastrophes, such as unprecedented global floods and massive, landscape-altering earthquakes.



LUCH VOLCANO: 15KM ASH BLAST & LIGH
KAMCHATKA, RUSSIA

Uniformitarianism

- Proposed in the 1700s by Scottish physician James Hutton, marking the true birth of modern geology.
- Fundamental Principle: **"The present is the key to the past."**
- It states that the physical, chemical, and biological laws operating today have also reliably operated throughout the geologic past.
- Features like deep canyons were not formed by sudden catastrophes, but by slow, steady erosion over millions of years.



Comparing the Paradigms

Catastrophism (Historical View)

- Earth is exceptionally young (thousands of years old).
- Geologic features formed instantly via violent, unrepeatable events.
- Focuses exclusively on sudden, global-scale disasters.
- Based heavily on early chronological calculations rather than field evidence.

Uniformitarianism (Modern View)

- Earth is incredibly old (billions of years old).
- Geologic features form gradually via continuous, steady processes.
- Focuses on observable, ongoing mechanisms (erosion, uplift, weathering).
- Serves as the absolute foundation of all modern geological sciences.

DISCUSSION

Let's talk about your questions and ideas.

Chapter References



Syllabus Primary Reference: Tarbuck, E.J., Lutgens, F.K., and Tasa, D.G. (2014). *Earth: An Introduction to Physical Geology* (11th ed.). Pearson Education, Inc., USA.



Supplementary Text: Plummer, C.C., Carlson, D.H., and Hammersley, L. (2015). *Physical Geology* (15th ed.). McGraw-Hill Education.